

Interfaces

1. Payment Interface

Task:

- Define an interface `Payment` with a method `processPayment(double amount)`.
- Create two classes `CreditCardPayment` and `PayPalPayment` that implement `Payment`.

Instructions:

- Each class should print how the payment is being processed.
- Write a main class to demonstrate both types of payments.

2. Flyable Interface

Task:

- Create an interface `Flyable` with methods `takeOff()`, `fly()`, and `land()`.
- Implement it in two classes: `Bird` and `Airplane`.

Instructions:

- Implement the methods with class-specific behavior.
- Use a polymorphic method to simulate a flying object in action.

3. Multiple Interfaces

Task:

- Define two interfaces:
 - `Readable` with `void read()`
 - `Writable` with `void write()`
- Create a class `Document` that implements both interfaces.

Instructions:

- Override both methods.
 - Create an object of `Document` and invoke both methods.
-

4. Interface with Constants

Task:

- Create an interface `VehicleConstants` that contains:
 - `MAX_SPEED`
 - `MIN_SPEED`
- Create a class `Car` that uses these constants to print allowed speed range.

Instructions:

- Show how the class accesses the interface constants.

5. Functional Interface and Lambda

Task (Java 8+):

- Create a functional interface `MathOperation` with a method `int operate(int a, int b)`
- Implement it using lambda expressions for:
 - Addition
 - Subtraction
 - Multiplication

Instructions:

- Write a method that accepts a `MathOperation` and two integers, then returns the result.
- Call this method with different lambdas.

6. Remote Control Interface

Task:

- Define an interface `RemoteControl` with methods:
 - `void turnOn()`
 - `void turnOff()`
 - `void changeChannel(int channel)`
- Implement it in two classes: `TV` and `SetTopBox`